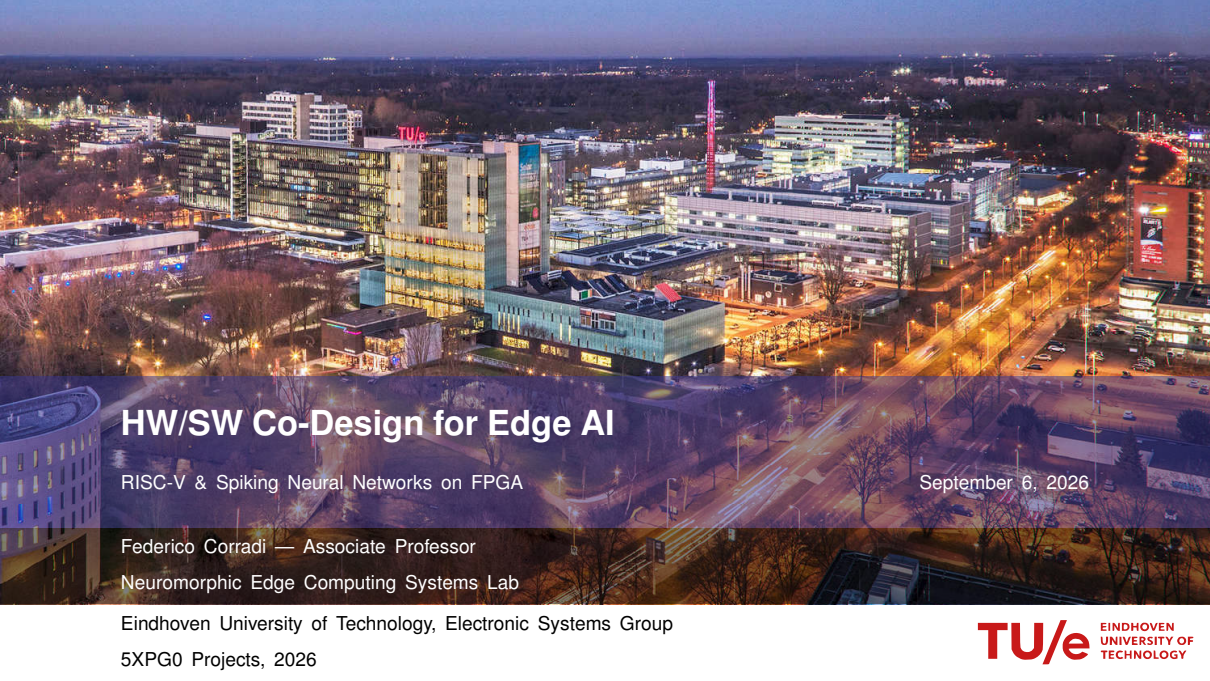




HW/SW Co-Design for Edge AI

RISC-V & Spiking Neural Networks on FPGA

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Federico Corradi — Associate Professor
Neuromorphic Edge Computing Systems Lab

Eindhoven University of Technology, Electronic Systems Group
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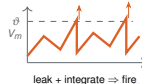
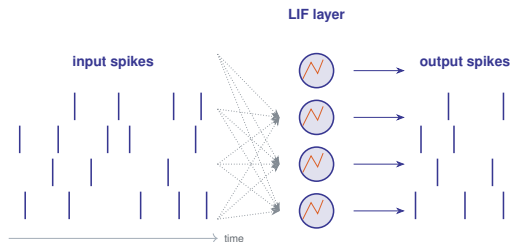
The project: a neuromorphic SoC on FPGA

Why

- Edge AI needs **efficient** processing architectures
- **Spiking Neural Networks (SNNs)** compute with sparse events in time — low power by construction
- CPUs are poor at running them: irregular, event-driven workloads

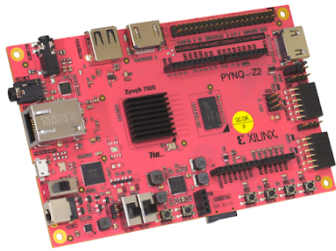
What you will do

- Design and evaluate a custom **System-on-Chip** for neuromorphic workloads
- Instantiate a soft-core **RISC-V** processor on the **PYNQ-Z2** FPGA
- Modify the system to run SNNs fast: from **custom ISA extensions** to **multi-core scaling**



Spiking Neural Network: layers of Leaky Integrate-and-Fire (LIF) neurons driven by events, not a clock.

Your platform: RISC-V on a PYNQ-Z2



PYNQ-Z2 (Zynq-7020, 512 MB DDR3)



open ISA: free to extend

The SoC you build

- Zynq-7000 SoC: ARM processing system (PS) + FPGA fabric (PL)
- PYNQ framework on the PS: Python-driven control of your PL design
- In the PL: an open-source **RISC-V soft core** (VexRiscv or PicoRV32)
- Around it: your own **SNN hardware accelerator**

Processing System (PS)

ARM Cortex-A9
Linux + PYNQ (Python)

AXI

FPGA fabric (PL)

RISC-V
soft core

SNN
accelerator

custom ISA extensions,
multi-core scaling

HW/SW co-design on the Zynq

Profile a software LIF model on the core → find the bottleneck → move it to hardware: **HW/SW co-design**.

Five-week plan, support & background reading

Weekly milestones (5 weeks, ~8 h/week)

- **W1–2** Baseline bring-up: RISC-V soft core on the FPGA; C/Asm benchmarks verify memory & peripherals
- **W3–4** Neuromorphic acceleration: profile the software LIF model, design accelerator / ISA extensions for spike generation & propagation
- **W5** Evaluation: latency, throughput, resource utilization

Supervision & prerequisites

- NECS-lab TA: two 30-min check-ins per week (Vivado / RISC-V toolchain guidance)
- **Required:** FPGA experience, working Vivado, VHDL/Verilog — advanced hands-on project

Publications (background reading)

- F. Corradi, G. Adriaans, S. Stuijk, *Gyro: a digital SNN architecture for multi-sensory data analytics*, 2021
<https://dl.acm.org/doi/pdf/10.1145/3444950.3444951>
- A. Sankaran, P. Detterer, K. Kannan, N. Alachiotis, F. Corradi, *An event-driven recurrent SNN architecture for efficient inference on FPGA*, ICONS 2022
<https://dl.acm.org/doi/pdf/10.1145/3546790.3546802>
- H. Irmak, F. Corradi, P. Detterer, N. Alachiotis, D. Ziener, *A dynamic reconfigurable architecture for hybrid spiking & convolutional FPGA-based NN designs*, JLPEA 2021
<https://www.mdpi.com/2079-9268/11/3/32>